

Mark Joseph Valenzuela

AI Workflow Automation Engineer

n8n • AI Automation • AI-Assisted Development • API Integration • Business Process Automation

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<https://markvalenzuela23.github.io/markjoseph-portfolio/> | github.com/markvalenzuela23

Professional Summary

IT professional with 4+ years of experience across systems administration, infrastructure, and technical support, now specialising in AI workflow automation and AI-assisted software development.

I build n8n workflows that integrate LLMs, OCR, messaging platforms, and Google Workspace to remove repetitive manual work, and I ship production software by directing AI coding agents — most recently coordinating six parallel Claude Code sessions to deliver a transaction management module inside a live SaaS platform.

I also run the infrastructure underneath what I build, including a zero-downtime migration of a production platform from Render onto a self-provisioned VPS running Dokploy, cutting hosting cost by roughly 86%.

Core Skills

AI & AUTOMATION

- n8n
- Workflow Automation
- AI Agents & Chatbots
- Prompt Engineering
- OpenAI API
- Google Gemini
- Business Process Automation

API INTEGRATION

- REST API • Webhooks
- HTTP Requests • JSON
- Google Sheets API
- Telegram Bot API
- Facebook Messenger API

SERVERS & DEPLOYMENT

- Dokploy • Docker • Docker Swarm
- Linux VPS provisioning (Hostinger KVM)
- PostgreSQL administration
- pg_dump • rclone offsite backups
- DNS • DMARC

AI-ASSISTED DEVELOPMENT

- Claude Code
- Parallel agent sessions
- Git worktree isolation
- React • Node.js • Express
- PostgreSQL • Drizzle ORM
- Test-backed delivery

AI PROCESSING

- OCR & Image Processing
- GPS Metadata Extraction
- Reverse Geocoding
- Knowledge Base Retrieval

INFRASTRUCTURE & PROGRAMMING

- Windows Server • Active Directory
- VMware ESXi • Veeam Backup
- Microsoft 365 • Exchange Online
- TrueNAS • Synology • pfSense
- Python • PowerShell • JavaScript
- Zabbix • Wazuh • Rundeck • iTop

Selected Projects

TRANSACTION MANAGEMENT MODULE — YAPII (AI REAL ESTATE SAAS)

TECHNOLOGIES: Claude Code, Node.js, Express, PostgreSQL, Drizzle ORM, React, Vite, Docker Swarm, Git Worktrees

- Rebuilt a brokerage's transaction management inside the company's AI SaaS platform to replace the paid third-party tool it depended on, matching it feature for feature.
- Directed six Claude Code sessions running in parallel, each isolated in its own git worktree and scratch database, coordinated into a single reviewable pull request.
- Built a trigger-based automation engine that creates tasks from file conditions, plus intake automation that polls live web forms and applies the matching task template.
- Delivered task management, reporting dashboards, and inbound email-to-listing matching across reply chains, forwards, and localised variants.
- Added 101 tests to code that previously could not be tested, and traced 236 suite failures to roughly five root causes.
- Enforced read-only access to every external system throughout the rebuild, with tests that fail if write paths are re-enabled.

PLATFORM MIGRATION — RENDER TO SELF-HOSTED VPS WITH DOKPLOY

TECHNOLOGIES: Dokploy, Hostinger KVM 8, Docker Swarm, PostgreSQL, pg_dump, rclone, DNS/DMARC

- Provisioned a Hostinger KVM 8 VPS from scratch and deployed Dokploy as a self-hosted alternative to Render, preserving push-to-deploy for the team.
- Reduced hosting from \$181 to \$25.64 per month — roughly \$1,860 per year — after evaluating and rejecting managed Postgres on cost.
- Completed the cutover with zero downtime by deploying to a spare hostname, verifying, then repointing the domain without restarting the container.
- Rehearsed the database migration to a 51-second restore of a 7.2 GB dump across 416 tables before cutting over.
- Fixed a Dockerfile that had never successfully built, and ended recurring hourly crashes by raising the Node heap limit.
- Replaced a paid backup add-on with scripted pg_dump and rclone to object storage at roughly \$1.50/month.

AI CUSTOMER SUPPORT CHATBOT

TECHNOLOGIES: n8n, Google Gemini, Facebook Messenger API, Google Docs, Webhooks, HTTP Requests, Prompt Engineering

- Designed and built an AI-powered support chatbot for a pisonet business, answering customer enquiries from a custom knowledge base without human intervention.
- Integrated Facebook Messenger with n8n and retrieved business information from a Google Docs knowledge base.
- Implemented automatic language detection to answer in English or Filipino.
- Engineered context-aware prompts so pricing and FAQ responses stayed accurate and consistent.

GCASH BUY LOAD AUTOMATION

TECHNOLOGIES: n8n, OpenAI API, Telegram Bot API, Google Sheets API, OCR, GPS Metadata Extraction, Reverse Geocoding, REST API

- Built an end-to-end workflow automating the GCash Buy Load reporting process for a reseller.
- Validated user licences through Google Sheets before processing any request.
- Accepted image uploads via a Telegram bot and extracted transaction details using OCR.
- Retrieved GPS coordinates from image metadata and converted them into human-readable addresses using AI.
- Recorded validated data into Google Sheets automatically and returned real-time confirmations through Telegram.

GCASH SUMMER NANO AUTOMATION

TECHNOLOGIES: n8n, Telegram Bot API, Google Sheets API, REST API, Webhooks, JSON

- Developed an automated Telegram-based reporting workflow with licence validation via Google Sheets.
- Automated collection, validation, and recording of report data, eliminating manual encoding.
- Sent instant confirmations and status updates, reducing human error during submission.

Professional Experience

TECHNICAL VIRTUAL ASSISTANT, PRODUCT DEVELOPMENT — HOMEXA

August 2026 – September 2026

- Worked on Yapii, the company's AI-powered SaaS platform for real estate agents (Node.js, PostgreSQL, React).
- Delivered the transaction management module by directing parallel Claude Code sessions to a single reviewable pull request.
- Migrated the platform off Render onto a self-provisioned Hostinger KVM 8 VPS running Dokploy, with zero downtime.
- Hardened destructive write paths to external systems and raised test coverage on previously untestable code.

IT INFRASTRUCTURE CONSULTANT — ST. CATHERINE DE NOVALICHES SCHOOL, INC.

2026 – Present

- Designed and deployed the school's complete network infrastructure supporting administrators, faculty, and students.
- Implemented a dual ISP setup with load balancing, structured cabling, network switches, and wireless access points.
- Planned cable routes and network topology, and provides ongoing maintenance and technical support.

WORKFLOW AUTOMATION ENGINEER — PRIVATE CLIENTS

2026 – Present

- Builds intelligent workflow automations using n8n integrated with Telegram, OCR, OpenAI, and Google Sheets.
- Delivered the GCash Buy Load and Summer Nano automations, including licence validation and automated confirmations.
- Designs AI chatbots that resolve customer enquiries from custom knowledge bases.

IT CONSULTANT — MR. FREEZE ICE PLANT

2026 – Present

- Provides on-call technical support, computer maintenance, and network administration for business operations.
- Handles Windows administration, printer support, system upgrades, and remote troubleshooting.

SYSTEM ADMINISTRATOR — WOODFIELDS CONSULTANTS, INC.

June 2023 – February 2026

- **Server & Network Management:** Provisioned and maintained physical and virtual servers, pfSense firewalls, network equipment, and NAS storage (TrueNAS, Synology).
- **Microsoft 365 & Active Directory:** Managed user accounts, licensing, Exchange Online, access rights, and AD domain migrations (ADMT).
- **Backup & Disaster Recovery:** Managed Veeam Backup & Replication for virtual machines.
- **Security & Monitoring:** Implemented Wazuh for threat hunting and Zabbix for infrastructure monitoring.
- **Service Desk:** Handled hardware and software support (SAP, QuickBooks) and analysed IT tickets in iTop.
- **Automation:** Automated repetitive administrative tasks using Python and Rundeck, and maintained IT documentation and SOPs.

FIELD SALES REPRESENTATIVE — KEYMOTORS INC.

September 2022 – June 2023

- Managed an assigned territory, monitored market conditions and competitors, and consistently met or exceeded personal quotas.

MARKETING ASSISTANT — PREMIUMBIKES CORP.

November 2020 – May 2022

- Produced marketing materials in Adobe Photoshop and Illustrator, and maintained the marketing database and lead lists.

IT ASSISTANT — SUPER GROCERS INC.

October 2019 – October 2020

- Installed and configured hardware and software, and tested new deployments before rollout.
- Diagnosed hardware, software, and network issues, configured employee workstations, and removed malware and ransomware.
- Maintained software licences and patched applications.

ACCOUNTING STAFF — SABROSA FOODS INC.

February 2019 – October 2019

- Maintained accounts payable records and processed invoices, journal vouchers, and reimbursements.

Education

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

STI College of Novaliches • June 2014 – May 2018

Automation Philosophy

I enjoy identifying repetitive manual tasks and turning them into reliable automated workflows. The goal is not simply to automate work, but to build solutions that save time, reduce human error, and let teams focus on higher-value activities — and to verify they actually run, rather than assuming a green build means a working feature.